

Case Study

Course Code	CDE2313	Course Name	Mobile Application Development
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TDM Insight

Native Android Therapeutic Drug Monitoring Calculator

Platform: Native Android | **Language:** Kotlin | **UI:** Jetpack Compose / XML | **Design:** Material 3

1. Case Study Background

A hospital pharmacy department performs Therapeutic Drug Monitoring (TDM) calculations for selected medicines. The calculations may require patient information, medication dose, dosing interval, drug concentration, sampling time, and laboratory information. The proposed TDM Insight application brings these inputs and calculations together in a structured native Android experience.

The concept is inspired by existing TDM calculators, including myTDM Calculator and the Malaysian Pharmacy Information System (PhIS) TDM Calculator documentation. The PhIS documentation describes Vancomycin Pre, Post, and Pre-Post workflows and associated pharmacokinetic outputs.

2. Project Scope

The project is deliberately focused. The main objective is to build a native Android TDM Calculator with a reliable, non-trivial calculation engine at its core. The mobile interface provides a clear way to collect inputs, validate them, run the calculation, and explain the results.

The core application should include:

1. Native Android development using Kotlin, Android Studio, and Jetpack Compose/XML.
2. Vancomycin TDM calculation workflows.
3. Dynamic input forms are based on the selected calculation method.
4. Input and cross-field validation.
5. A dedicated calculation engine separated from the user interface.
6. Intermediate and final pharmacokinetic results.
7. A clear explanation of how the main results were obtained.
8. A clean and intuitive Material 3 user interface.

The following are not required for the core project:

1. User authentication.
2. Cloud backend or online services.
3. Analytics dashboards.
4. Complex patient-management systems.
5. Multiple medication modules.
6. Large educational-content modules.

3. Main Calculation Workflows

The core application should focus on three Vancomycin workflows:

Vancomycin Pre: A workflow based on pre-dose concentration.

Vancomycin Post: A workflow based on a post-dose concentration and its sampling information.

Vancomycin Pre + Post: A workflow using both pre-dose and post-dose concentrations and relevant timing information.

The selected workflow should determine which fields are shown. Students should not display every possible input field on one screen.

4. Example User Flow

1. Open TDM Insight.
2. Create a fictional case.
3. Enter the required patient parameters.
4. Select Vancomycin.
5. Select Pre, Post, or Pre + Post.
6. Enter the values required by the selected workflow.
7. Validate the information.
8. Review the calculation inputs.
9. Run the TDM calculation.

10. Display intermediate results.
11. Open the calculation explanation.

5. Calculation Engine

The calculation engine is the application's main technical component. It should receive structured input data, validate it, run the appropriate calculation pathway, and return structured results. Do not embed calculation logic directly in Composable or UI functions.

Conceptually:

UI → Input State → Validation → TDM Calculation Engine → Result Model → Results UI

6. Pharmacokinetic Outputs

Depending on the selected workflow and lecturer-approved calculation specification, the application may calculate:

1. Elimination rate constant (K_e).
2. Elimination half-life.
3. Volume of distribution (V_d).
4. Clearance.
5. Concentration-related values.
6. AUC-related values.
7. Other parameters are required by the selected TDM method.

The exact clinical equations, units, assumptions, and reference values must be based on lecturer-approved authoritative sources. Students must not invent clinical formulas.

7. Dynamic Forms

The input screen should adapt to the selected calculation. For example, a Pre + Post workflow may require both concentration values and additional sampling-time information, while a Pre workflow may require a different set of inputs.

8. Validation

Validation should go beyond checking whether a field is empty.

1. Required-field validation.
2. Numeric and unit validation.
3. Range validation where specified.
4. Crossfield validation, such as logical timing relationships.
5. Detection of missing values required by the selected workflow.

6. Protection against mathematical errors such as division by zero or invalid logarithmic operations.
7. Clear messages that distinguish an error from information that simply requires review.

9. Explainable Results

The result screen should not display only a final number. Users should be able to open a calculation explanation that shows key input values, intermediate calculations, and the result in a simplified sequence.

Input Values → Intermediate Values → Pharmacokinetic Parameters → Final Result

10. Optional Enhancements

The following features may be added to enhance the application but are not required for the core scope:

1. Local calculation history.
2. Camera capture of a fictional laboratory report or medication label.
3. OCR-assisted extraction of a selected value, with mandatory user confirmation.
4. What-if or scenario simulation.
5. Additional TDM medication module.
6. Simple concentration-time graph.
7. Export or sharing of a calculation summary.

11. Camera Feature — Optional Example

If the camera enhancement is selected, the camera should have a meaningful role in the TDM workflow. A suitable example is capturing a fictional laboratory report and attaching the image to the calculation. An advanced version may use OCR to help extract a selected value. Users must always review and confirm extracted values before using them.

Camera → Capture → Review → Confirm Value → Use in Calculation

13. Reference Application and Clinical Sources

Students should use the following as starting points for domain understanding and reference checking:

myTDM Calculator: <https://www.mytdmcalculator.com/>

PhIS TDM Calculator Manual: Malaysian Pharmacy Information System documentation covering multiple TDM workflows.

Vancomycin TDM Guidance: Consult current authoritative clinical guidance for any clinical equation or target used in the application.

14. Important Clinical Disclaimer

TDM Insight is an academic software prototype. It is intended only for educational and software development purposes. It must not be presented as a clinically validated prescribing, diagnostic, or autonomous treatment-decision system. All demonstration cases must be fictional, and clinical equations and reference values must be supported by appropriate authoritative sources.

15. The Main Challenge

The objective is not simply to build a calculator screen. The challenge is to turn a non-trivial TDM calculation workflow into a reliable, understandable mobile application.

Real-World TDM Problem → Dynamic Mobile Input → Validation → Calculation Engine → Explainable Results